

VALAY SAITWADEKAR

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EDUCATION

California State University Fullerton, United States of America
Master of Science in Computer Science

May 2026
CGPA 3.95/4

Shree L. R. Tiwari College of Engineering, Mumbai University, India
Bachelor of Engineering in Computer Engineering with
Honors in Cyber Security with *Distinction*

May 2024
CGPA 9.11/10
(3.65/4)

PROFESSIONAL EXPERIENCE

Filtroll International (India) Pvt. Ltd., Mumbai, India

December 2022 – June 2023

Intern – Software Engineer

- Designed and shipped full-stack inventory, accounting and CRM modules using Python and Flask, applying object-oriented design principles, data structures and algorithmic problem solving to enforce data integrity and business logic, reducing manual work by 50% and improving inventory accuracy by 85%.
- Collaborated with cross-functional teams in fast-paced short development cycles, gathering requirements and delivering proprietary software releases incrementally with clear written and verbal communication and measurable outcomes.
- Designed a relational database schema in MySQL to organize business data for reports and analysis with stored procedures and database triggers to enforce business logic and data integrity.
- Created user guides and admin manuals to support onboarding new users and documentation for users covering system navigation and module functions and conducted training to help employees adopt the new ERP tools.

PROJECTS

DupliCron – Metadata-Based Incremental File Synchronization and Backup

- Built a cross-platform C++20 high-performance file synchronization utility for Linux and Windows, applying OOP design, low latency data structures and algorithmic problem solving to implement metadata-based change detection and a binary cache, validated against 1M+ files with memory footprint in the low MBs.
- Applied deep C++ memory management techniques, cache-conscious data structure design and memory layout optimization, minimizing heap overhead and maximizing sequential access throughput across large file sets with sub-millisecond synchronization.
- Architected a storage-aware copy pipeline with runtime-configurable I/O strategies, applying computational optimization and algorithmic design to maximize throughput across storage subsystems accounting for hardware behavior and storage hierarchy for optimal latency.
- Implemented multi-threaded C++ system programming with a custom thread pool, thread-safe queues supporting serialized and parallel modes, overlapping scanning, hashing and copying phases to maximize CPU and disk utilization, enforcing data integrity via post-copy verification and failure-aware metadata updates with automatic recovery.

TransactViaMe – Event-Driven Payment Backend Platform

- Built and orchestrated distributed real-time payment workflows across authorization, settlement and ledger services using Java and Python, applying OOP design patterns and data structures connected via Kafka, PostgreSQL and Redis, exposing low-latency functionality through REST APIs serving real-time decisions at scale.
- Implemented high-throughput distributed pipelines processing 50K+ payment events per minute, applying algorithmic optimization and multi-threaded design with idempotent consumers, retries and failure recovery ensuring data integrity and system reliability during restarts and load spikes.
- Enabled automatic horizontal scaling on AWS EKS using Kubernetes with Horizontal Pod Autoscaler, maintaining API response times under 200ms under synthetic stress testing with k6/Locust, validating distributed system behavior across pod restarts and consumer rebalancing.
- Developed supporting TypeScript services and risk-monitoring tooling, validated system reliability through stress and failure testing, and maintained full observability via Prometheus, Grafana and centralized logging across high-volume distributed production traffic.

TECHNICAL SKILLS

- Programming Languages: C, C++ (keeping pace with C++26), Java, Kotlin, Python, SQL, R, Shell Scripting, HTML, CSS, JavaScript, TypeScript
- C++ Core: Memory Model, RAII, Copy/Move Semantics, Smart Pointers, Perfect Forwarding, Value Categories, STL Internals, Custom Allocators, VTable Internals, Templates & Metaprogramming, SFINAE, Type Traits, Lambdas, Structured Bindings, Inline Variables, ADL, Object Lifetime, Storage Duration, Multiple Inheritance, EBO, cv-qualifiers
- Concurrency: Lock-free Programming, Wait-free Programming, Atomics, Memory Ordering, False Sharing, Memory Barriers, Hazard Pointers, RCU, Thread Pools, Task Queues, Synchronization Primitives, Futex, Concurrent Data Structures, Multithreading, Thread Local Storage, Double Buffering, Condition Variables
- CPU & System Architecture: Cache Hierarchy, Out-of-order Execution, Branch Prediction, x86 Microarchitecture, CPU Pipelining, NUMA Topology, Virtual Memory, SIMD, Prefetching, Instruction Level Parallelism, Superscalar Execution, Hyperthreading, Write/Store Buffers, OS Scheduling, Kernel and User Space, Memory Management, Clock Gating
- Hardware & System Tuning: Kernel Bypass, CPU/IRQ Affinity, Huge Pages, DMA, Cache Warming, Memory Locking, C/P-State Management
- Low Latency & Performance: Hot Path Optimization, Cache Line Alignment, Branchless Coding, Zero Copy Design, Inlining, Vectorization, Loop Unrolling, Slow Path Removal, Lock-free Data Structures, Cache Conscious Data Structures, Aligned Pointers, RVO/NRVO, Memory Footprint Reduction
- Debugging & Profiling: perf, Flamegraphs, KCachegrind, Callgrind, Google Benchmark, Valgrind, gdb, gprof, strace, x64dbg, WinDbg
- Build & Tools: CMake, Make, Bazel, GTest, Git, GitHub, CI/CD, Jenkins, Gradle, Maven, Test Driven Development, Unit & Integration Testing
- Distributed Systems: AWS, Docker, Kubernetes, Prometheus, Grafana, Kafka, event-driven architecture, real-time systems, fault tolerance
- Frameworks & Tools: Flask, FastAPI, Spring Boot, Node.js, REST APIs, PostgreSQL, MySQL, Redis, Postman, k6, Locus
- Collaboration & Process: Jira, Agile, code reviews, cross-functional collaboration, GitHub Copilot, Claude Code, AI-assisted development

RELEVANT COURSEWORK

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|--------------------------------|--|-------------------------|
| Operating Systems Design | System Programming & Compiler Construction | Microprocessor |
| Advanced Computer Networking | Digital Logic & Computer Architecture | Distributed Computing |
| Software Design & Architecture | Theory of Computer Science | Adv. Algorithm Analysis |